

Thanh Khoi Nguyen

Ho Chi Minh City | nguyenthankhoi123x321@gmail.com | +84 768 953 579 | 0009-0003-5879-451X
linkedin.com/in/khoi-nguyen-thanh | github.com/NTKHarry

Education

University of Science - VNUHCM, Honor Programme in Computer Science Sep 2023 - May 2027

- GPA: 3.94/4.0 (**Top 10** in programme); Dean's Honor List **Top 3** (2024), **Top 2** (2025)
- Admitted to the Information Technology Talent Program (**top 45** students in faculty)

Experience

Research Student, Saigon AI Hub (SAIH), Applied AI Lab Early 2026 - Present

- One of 10 inaugural research groups at Vietnam's first open AI hub, selected for Google Labs' AI Futures Fund; focus on multimodal sport reasoning and video-language understanding.

Research Student, Software Engineering Lab (SELab), HCMUS Jun 2024 - Present

- Research spanning 3D vision, multimodal reasoning and generation.
- **8 papers** (as first, co-first, and co-author): 1 journal (Q2, *accepted*); 1 CORE A* (*accepted*), 2 CORE A* (*under review*); 2 CORE C (*accepted*); 1 CORE B main conference (*under review*); 1 CORE B workshop (*accepted*); 1 Scopus-indexed conference (*accepted*).

Projects

AgentSys - Multimodal Media Agent May 2026 - Present

- Built an agentic system that goes beyond retrieval to take action over personal media - grouping, querying, and reasoning over image libraries through natural-language interaction.
- Deployed and served **Florence-2** as a self-hosted inference server to power the agent's grouping/querying pipeline, decoupling model inference from agent logic.
- Awarded **Best Project** at Smart Software Innovation Program (SSIP) 2026, collaboration with VNG Group.

Soccer Video Analysis Agent Apr 2026 - Present

- Developed an agent-based system for structured visual question answering over soccer broadcast video, submitted to the **SoccerNet 2026 VQA Track (top 5)**.
- Served the underlying LLM using **vLLM** for low-latency inference in a real-time video QA setting.
- Adapted the same agent for the **ACM MM 2026 TRIDENT Grand Challenge** (deepfake detection workshop), achieving **top 3** in the video track.

MediaEval 2025 - News Thumbnail Generation May 2025 - Oct 2025

- Built a data pipeline to crawl and clean large-scale newspaper article data as structured input for LLM-based content understanding.
- Engineered an end-to-end multimodal pipeline combining LLM content understanding with diffusion-based image synthesis; ranked **top 3 worldwide (9% above baseline)**.
- Released **NewsGen**, a dataset of **6,000** labeled thumbnail images, to advance open research in automated thumbnail generation.

Course Projects (all awarded 10/10) Nov 2025 - Dec 2025

- **VieLink**: Co-engineered a full-stack link-in-bio platform with real-time username validation, customizable theming, and an LLM-powered bio writer; deployed end-to-end from concept within one semester.
- **AI-Powered Nutrition Tracking App**: Architected an Android application integrating **Gemini** and **ML Kit** for on-device food recognition via camera and real-time nutritional analysis.

Technologies

Languages: Python (primary), SQL, C++, C, Java, R, MATLAB

ML/AI: PyTorch (primary), TensorFlow, OpenAI API, Anthropic SDK, LangChain, CrewAI, Pydantic AI, MCP, RAG & vector search (Qdrant), Vision-Language Models (CLIP, TinyCLIP, Florence-2), prompt engineering (few-shot, CoT)

MLOps/Deployment: Docker, vLLM (LLM serving), CI/CD basics, Git

Data: Web crawling/scraping (Crawl4AI, BeautifulSoup), data cleaning, pandas, numpy, MySQL, PostgreSQL

Skills

Core: Scientific Research, Technical Writing, Problem Solving, Programming Logic, Machine Learning, Computer Vision

Languages: Vietnamese (native), English - Professional working proficiency (IELTS 7.0)